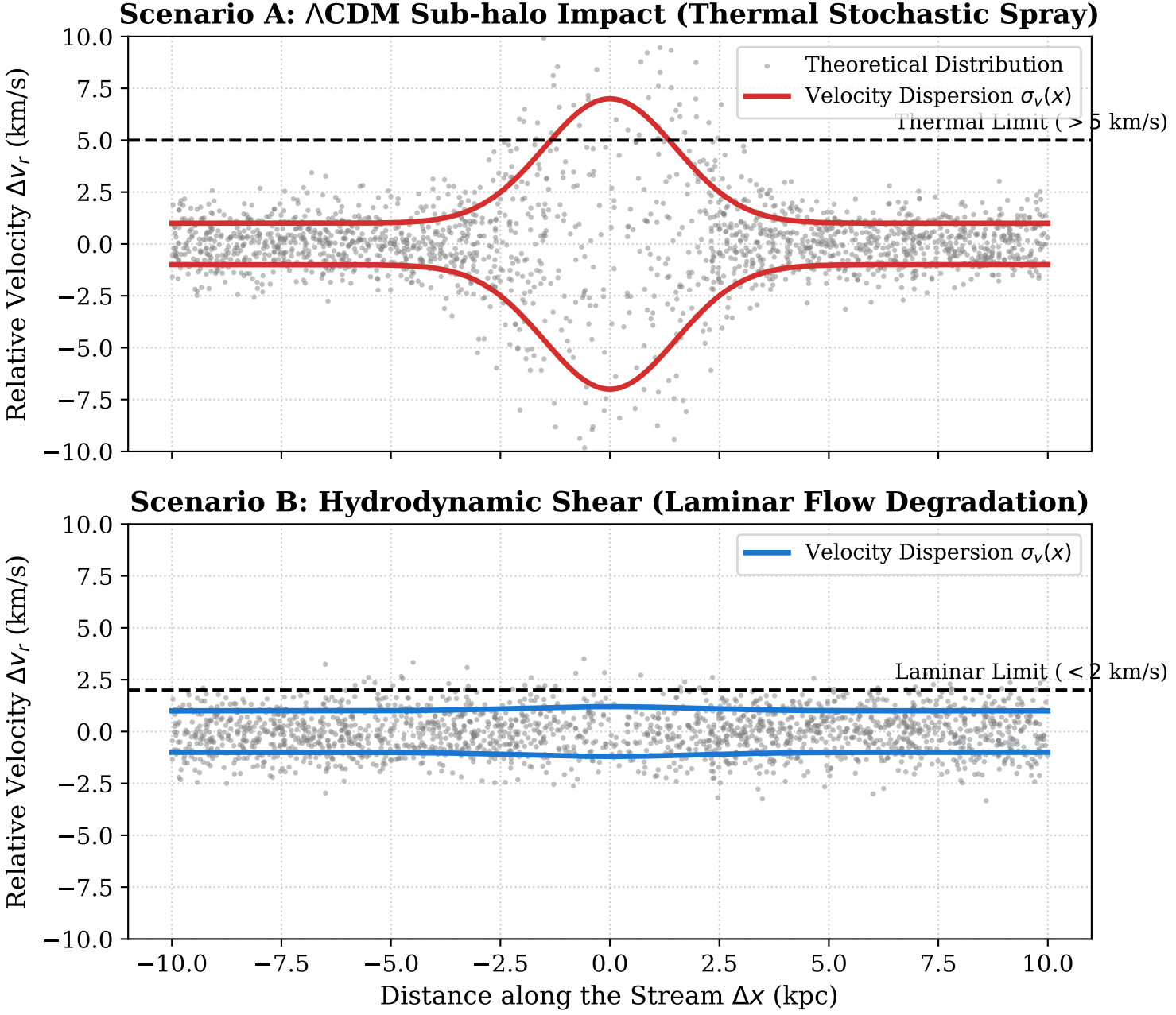

Scenario A: ΛCDM Sub-halo Impact (Thermal Stochastic Spray)
Theoretical Distribution
Velocity Dispersion $\sigma_v(x)$
Thermal Limit (> 5 km/s)
Relative Velocity Δv_r (km/s)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Scenario B: Hydrodynamic Shear (Laminar Flow Degradation)
Velocity Dispersion $\sigma_v(x)$
Laminar Limit (< 2 km/s)
Relative Velocity Δv_r (km/s)
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Distance along the Stream Δx (kpc)